ELSEVIER

Precambrian Research 69 (1994) 81–93

Precambrian Research

Palaeomagnetism of the Late Precambrian Vadsø and Barents Sea Groups, Varanger Peninsula, Norway

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Received July 13, 1993; revised version accepted January 29, 1994

Abstract

A palaeomagnetic study has been made on Neoproterozoic rocks from the Varanger Peninsula, northernmost Norway. Several of the formations studied contain a stable magnetization of diagenetic origin. Two main groups of directions were obtained; one (A) with a steep southeasterly positive direction, the other (B) with a shallow northern direction. The Group A direction is not in accordance with the stratigraphic age of the rocks and the Fennoscandian apparent polar wander path; hence it may reflect a remagnetization. The Group B direction yields a Late Riphean–Early Vendian palaeopole at 24°N, 207°E ($d_p=5°$, $d_m=11°$, $N=11$) in good agreement with data from contemporary rocks of the Kola Peninsula.

The Varanger Peninsula is transected by the Trollfjord–Komagelv Fault which divides the peninsula into the allochthonous Barents Sea Region and the autochthonous Varangerfjord Region. Previous palaeomagnetic studies have suggested that the Barents Sea Region has moved 500 to 1000 km to its present position. This study indicates that the movement has been less than 250 km which is in better agreement with geological information.

1. Introduction

The Neoproterozoic (Late Riphean and Vendian) period of the Earth's history involved the formation and split of large landmasses. Recently, new ideas about the ancient configuration of these blocks have emerged (e.g. Hoffman, 1991) and palaeomagnetism is an important method for tracing these movements. However, palaeomagnetic data are scarce for this time period, particularly for the Fennoscandian Shield. The rock units available on the Shield are sedimentary successions within and along the eastern margin of the Scandinavian Caledonides, often affected by this orogeny and the Late Riphean–Early Vendian autochthonous Visingsö Beds of southern Sweden. The latter are the subject of a separate study. This paper deals with palaeomagnetic data from the autochthonous Late Riphean–Early Vendian Vadsø Group on the southern part of the Varanger Peninsula and the contemporary allochthonous Barents Sea Group in the northern part of the Varanger Peninsula.

2. General geology

The geology of the Varanger Peninsula, East Finnmark has been studied over the last few decades and the stratigraphy and geological history is well understood (e.g. Banks and Røe, 1974; Banks et al., 1974; Føyn 1985; Siedlecka et al., 1989). The Geological Survey of Norway has

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SSDI 0301-9268(94)00015-J

published a geological map of the Varanger Peninsula (Siedlecki, 1980) and Hambrey (1988) and Nystuen and Siedlecka (1988) have presented an overview of the Neoproterozoic geology of the area.

The southern part of the peninsula, the Varangerfjord Region (Fig. 1) is underlain by an autochthonous succession of Late Riphean–Vendian sedimentary rocks divided into three units, in ascending order: the Vadsø Group, the Tanafjord Group and the Vestertana Group. These units are separated from the allochthonous rock succession in the Barents Sea Region in the north by the Trollfjord–Komagelv Fault (Johnson et al., 1978). The Barents Sea successions comprise the Barents Sea Group and the Lökvikfjell Group. Furthermore, the Caledonian Front with the Kalak Nappe Complex covers the northwestern corner of the peninsula (Fig. 1). The formations of the Vadsø Group and the Barents Sea Group and their lithologies are presented in Table 1.

The Vadsø Group rests unconformably on Precambrian gneisses while its uppermost formation, the Ekkerøy Formation, is conformably overlain by the Grønnes Formation of the Tan-

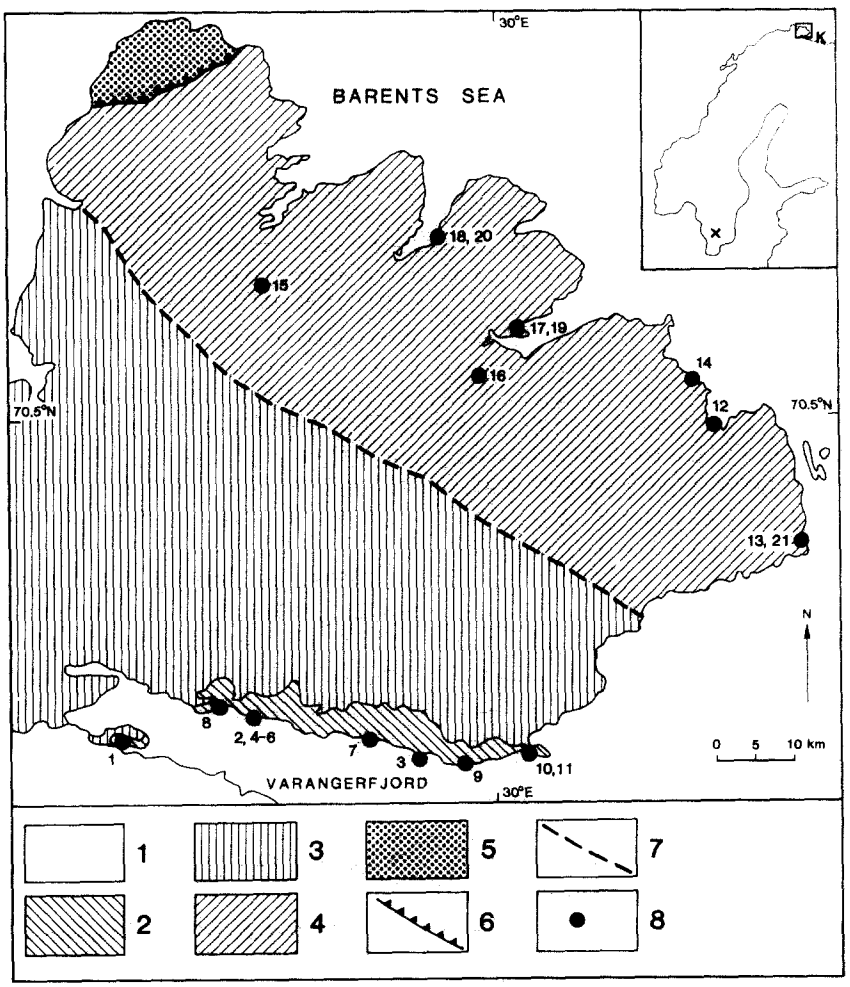


Fig. 1. Simplified geological sketch map of the Varanger Peninsula. *1*=Precambrian basement, *2*=Vadsø Group, *3*=Tanafjord and Vestertana Groups, *4*=Barents Sea Region, *5*=Caledonian nappes, *6*=Caledonian Front, *7*=Trollfjord–Komagelv Fault, *8*=palaeomagnetic sites. *K* and × (inset) denote the Kola and Visingsö areas, respectively.

Table 1
Formations and lithologies of the Vadsö and Barents Sea Groups (modified after Nystuen and Siedlecka, 1988)

Vadsø Group	
Ekkerøy Fm., 15–190 m	Shallow marine shale and sandstone
Golneselv Fm., 50–135 m	Fluvial sandstone
Paddeby Fm., 25–120 m	Fluvial sandstone
Andersby Fm., 25–40 m	Deltaic and lacustrine shale and mudstone
Fugleberg Fm., 125 m	Fluvial sandstone
Klubbnes Fm.	Deltaic shale
Veidnesbotn Fm., 300 m	Shallow marine and fluvial sandstone
Barents Sea Group	
Tyvjofjell Fm., 1500 m	Shallow marine and fluvial sandstone
Båtsfjord Fm., 1400–1600 m	Shallow marine sandstone, mudstone, limestone, dolomite
Båsnæring Fm., 2500–3500 m	Deltaic shale, siltstone and sandstone
Kongsfjord Fm., thickness unknown	Submarine sandstone and shale

afjord Group. The sedimentary sequences along the northern shore of the Varangerfjord are little affected by tectonic deformations and are gently folded and dip up to ten degrees in a northerly direction. Microfossils (acritarchs) from siltstones and mudstones of the Vadsø Group are well preserved and their yellow to brown colours indicate a mild post-diagenetic temperature regime (Vidal, 1981). Rice et al. (1989) show that the metamorphic grade within the autochthon of the East Finnmark–Varanger Peninsula varies from uppermost anchizone grade close to the Caledonian Front to low diagenetic grade farther east.

The rocks of the Barents Sea Region are divided into the Barents Sea Group and the Løkvikfjell Group. The thickness of the Barents Sea Group is estimated to be more than 9000 m, the base being unknown. The Barents Sea Group is transgressively overlain by the terrigenous Løkvikfjell Group with an low-angle unconformity. Tectonic deformation of the Barents Sea Group is variable but tends to increase towards the Caledonian Front in the northwest. Acritarch assemblages show variability in the preservation and colour of organic material, thus indicating local variations in thermal alteration (Vidal and Siedlecka, 1983). Rice et al., (1989) have shown that the Barents Sea Region rocks have reached epizone-grade metamorphism. The region is transected by swarms of NE-trending dolerite dykes, the Kongsfjord and Båtsfjord dykes.

The Trollfjord–Komagelv Fault has been considered as a major structural boundary separating the Barents Sea Region from the autochthonous foreland south of the fault. Based on palaeomagnetic data on the 640 Ma Båtsfjord dykes a dextral offset of 500–1000 km (Kjøde et al., 1978) or a sinistral offset of 500 km (Pesonen et al., 1989) have been suggested. Based on stratigraphic correlations across the fault Townsend et al. (1990) and Rice and Townsend (1991) have proposed only minor movements along the fault, perhaps 50 ± 50 km.

3. Age relationships

The Klubbnes Formation of the Vadsø Group has yielded an age of 810 ± 90 Ma (Rb–Sr whole rock; Pringle, 1973) and the Nyborg Formation of the overlaying Vestertana Group has been dated to 653 ± 23 Ma (Rb–Sr whole rock; Pringle, 1973). Dallmeyer and Reuter (1989) have dated the cleavage of the Nyborg Formation at ~635 Ma using the ^{40}Ar–^{39}Ar method (whole rock). Rb–Sr whole-rock dating of cleaved mudstones of the Kongsfjord Formation gave an isochron age of 520 ± 47 Ma for the development of cleavage (Taylor and Pickering, 1981). Vidal (1981) and Vidal and Siedlecka (1983) suggested biostratigraphic correlations based on acritarchs from the Vadsø and Barents Sea Groups. They considered that the Vadsø Group is Upper Riphean with a transition to the Vendian within the Ekkerøy Formation and that the Barents Sea Group is Upper Riphean with a transition to the Vendian within the Båtsfjord Formation.

K–Ar age determinations on the dolerite dykes of the Varanger Peninsula (Beckinsale et al., 1975) display three age groups, one group with

ages of more than 1000 Ma and a second of 640 Ma. Both groups are restricted to the Barents Sea Region. The third group comprises dykes with an age of ~340 Ma. This group occurs on both sides of the Trollfjord–Komagelv Fault.

4. Palaeomagnetic study

4.1. Sampling and measurement

Samples were collected from all formations of the Vadsø Group with the exception of the Paddeby Formation. In the Barents Sea Group, the Kongsfjord, Båsnæring and Båtsfjord Formations were sampled. Only hand-samples were collected. In-situ orientation of the samples was made with magnetic and sun compasses. From each sample up to five specimens were drilled and cut for measurements. Measurements were made on Geofyzika JR-3 and JR-5 and on Digico and Molspin fluxgate spinner magnetometers. Conventional stepwise alternating field (AF) and thermal demagnetization techniques were used for isolating the characteristic magnetic components. Chemical demagnetization was made using 8 *N* hydrochloric acid. Holes were drilled in the samples to facilitate the penetration of the acid. The solution time was up to 1800 hours.

In Table 2 are listed the number of samples (N) and specimens (n) that yielded characteristic magnetization used for calculations. Some sites yielded only one or two samples with a characteristic magnetization. To use these data is statistically improper but they are still included as the results obtained are considered of value for the discussion.

Often a substantial increase in magnetic intensity was observed during the later steps of the thermal demagnetization indicating changes in magnetic mineralogy; these alterations were confirmed by susceptibility measurements.

4.2. The Vadsø Group

The collection of rocks from the Vadsø Group comprises grey and red sandstones and siltstones. Generally, the samples with a stable magnetization presented a single characteristic magnetic component. Figs. 2 and 4 demonstrate typical demagnetization behaviour for the Fugleberget and Ekkerøy Formations (sites 3 and 11). The AF and thermal demagnetization showed a high coercivity and high blocking temperature component indicating haematite as the main remanence carrier at site 3. The corresponding data for site 11 indicate magnetite as the carrier of the characteristic remanence. Microscopic investigations showed the presence of limonite and pyrite. During thermal demagnetization these minerals are transformed into secondary haematite and magnetite, which acquire a hard thermochemical remanent magnetization that can be difficult to remove. An example is given in Fig. 3 where the increase in magnetic moment observed after heating to 400°C probably shows the formation of secondary magnetite. A corresponding increase was observed in the bulk susceptibility.

4.3. Barents Sea Group

Red sandstones were collected from the Båsnæring Formation and red mudstones from the Båtsfjord Formation. Both units displayed similar behaviour during demagnetization and typical examples are presented in Figs. 5 and 6. The AF demagnetization displays a high coercivity component and the thermal demagnetization indicates that this is carried by haematite, probably the fine haematite pigment that colours the rock. In addition, the slope and upturn of the thermal demagnetization curve (Fig. 5) indicate both the presence of secondary magnetic minerals with lower blocking temperature than haematite and the formation of new magnetic phases during heating. Specimens from eleven samples of the Kongsfjord Fm. were measured. They did not reveal any characteristic magnetic directions.

The chemical demagnetization tests gave the same directions as the AF- and thermal-demagnetization tests. Examples are given in Figs. 6c and 7.

Table 2
Palaeomagnetic directions

Site, Formation	n/N	Decl. (°)	Incl. (°)	α_{95} (°)	k	Group
Vadsø Group						
1. Veidnesbotn Fm.	6/4	299	−33	12.3	21.7	
2. Klubbnes Fm.	5/3	355	−8	15.7	15.9	B
3. Fugleberg Fm.	8/5	17	1	7.6	33.9	B
4. Fugleberg Fm.	7/4	1	27	12.5	18	B
5. Fugleberg Fm.	8/5	67	82	9.2	29.1	
6. Fugleberg Fm.	8/3	73	25	16.1	9.5	A
						A
7. Andersby Fm.	12/5	348	71	14.4	10.4	A
8. Andersby Fm.	10/6	12	46	14.8	9.9	B
9. Golneselv Fm.	4/3	156	63	20.8	11.3	A
10. Golneselv Fm.	5/3	157	80	10.6	35.1	A
11. Ekkerøy Fm.	21/14	167	67	6.2	24.6	A
Barents Sea Group						
12. Båsnæring Fm.	20/10	6	10	6.8	20.9	B
13. Båsnæring Fm.	20/12	2	0	6.2	25.2	B
14. Båsnæring Fm.	1/1	5	4	–	–	B
15. Båsnæring Fm.	5/3	1	26	13.2	22.4	B
16. Båsnæring Fm.	1/1	358	5	–	–	B
17. Båsnæring Fm.	4/2	16	−15	25.3	7.7	B
18. Båsnæring Fm.	5/2	341	−3	12.8	24	B
Mean Båsnæring Fm.	7	2	4	10.9	23.6	
19. Båtsfjord Fm.	11/4	95	73	13.3	10	A
20. Båtsfjord Fm.	1/1	99	53	–	–	A
21. Båtsfjord Fm.	7/4	353	23	18.8	7.9	B
Mean group B	11	1	6	8.9	22.5	B

n=number of specimens, N=number of samples from each site that gave characteristic directions. Mean directions are based on specimen data. Decl. and Incl.=site mean declination and inclination values. α_{95}=the semi-angle of the 95% cone of confidence. k=best estimate of Fisher's (1953) precision parameter.

5. Palaeomagnetic directions

The directions of the characteristic magnetic components were obtained by using a least-squares algorithm (Torsvik, 1986) combined with analyses of stereograms. Site mean directions for both the Vadsø and the Barents Sea Group formations are given in Fig. 8. The data shown are those obtained after tilt correction.

5.1. Vadsø Group

Reddish sandstones were sampled close to the base of the Veidnesbotn Formation, the lowermost unit of the Vadsø Group. The northwesterly declination and intermediate negative inclination separate it from the main directions obtained for the Vadsø Group.

Grey siltstones were collected in the upper part of the Klubbnes Formation close to the border of

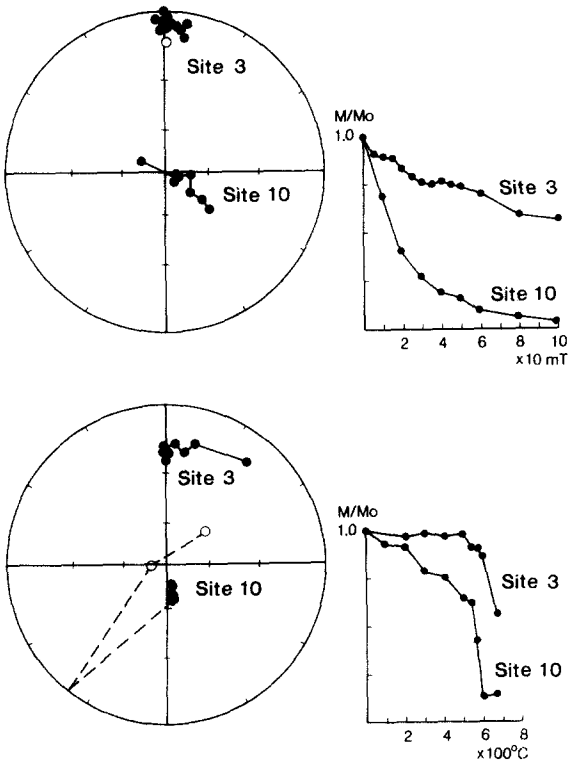


Fig. 2. Examples of behaviour of specimens from the Vadsø Group during alternating field (top) and thermal (bottom) demagnetization. Left: stereograms; closed (open) symbols denote positive (negative) inclination. The stereograms refer to the palaeohorizontal. Right: normalized demagnetization curves.

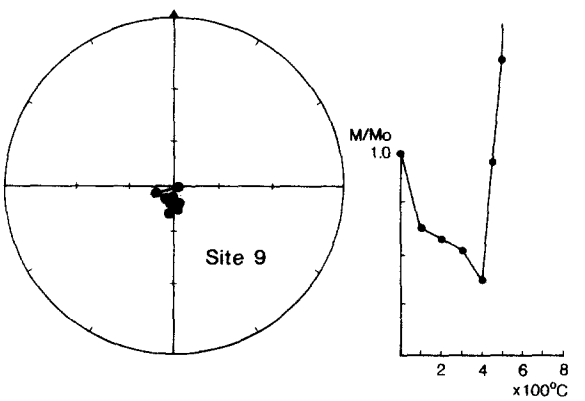


Fig. 3. Example of an anomalous increase in magnetic moment during thermal demagnetization due to changes in magnetic mineralogy during heating. Conventions as in Fig. 2.

the Fugleberget Formation. A shallow northerly direction with negative inclination was obtained from three samples.

The Fugleberget Formation was sampled at two sites. A grey sandstone from the upper part of the formation was collected at site 3 (Fig. 1). Five samples presented a hard stable magnetization with shallow northerly direction. Grey and red sandstones were collected at the second site. There, three different magnetic components, represented by 4–6 in Figs. 1 and 8, were isolated from twelve samples (Table 2). One direction (4) can be compared with the direction obtained at site 3. It was found in samples from the lower part of the formation close to the contact with the Klubbnes Formation. 5 and 6 are from the middle part of the formation and they yield steep to intermediate positive directions pointing east.

Grey and reddish siltstones and sandstones of the Andersby Formation were sampled at sites 7 and 8 (Figs. 1 and 8). Site 7 presented a steep northerly positive direction and site 8 an intermediate positive northerly direction.

The Golneselv Formation was sampled at sites 9 and 10 (Figs. 1 and 8). The grey sandstone of site 9 gave a steep positive southeasterly direction similar to the directions obtained from red and grey sandstones of site 10. Site 10 was sampled just below the basal conglomerate of the overlying Ekkerøy Formation, while site 9 was collected in the lower member of the Golneselv Formation.

Samples of red and grey siltstone and sandstone from the Ekkerøy Formation, site 11, yielded a steep southeasterly positive direction (Figs. 1 and 8). The direction is similar to that of the underlying Golneselv Formation.

5.2. The Barents Sea Group

No stable characteristic magnetization was found in the Kongsfjord Formation, the lowermost unit in the Barents Sea Group.

The Båsnæring Formation was sampled at sites 12 to 18 (Figs. 1 and 8). At site 15, grey siltstones were collected in the lower part of the formation, in the Næringselva Member. At the other

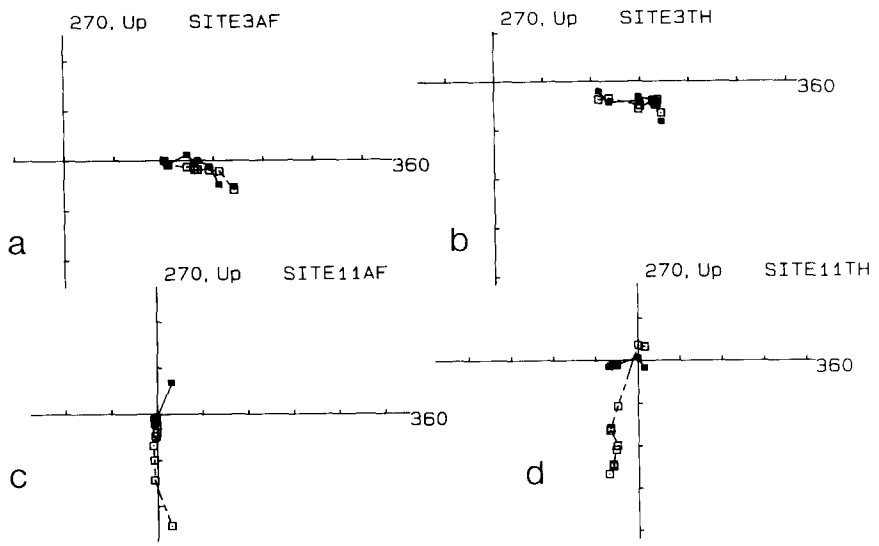


Fig. 4. Orthogonal vector projections of specimens from the Vadsø Group. Open (closed) symbols denote points in the vertical (horizontal) plane. Units on axis (a) and (b), 5 mA/m, on (c) and (d), 10 mA/m.

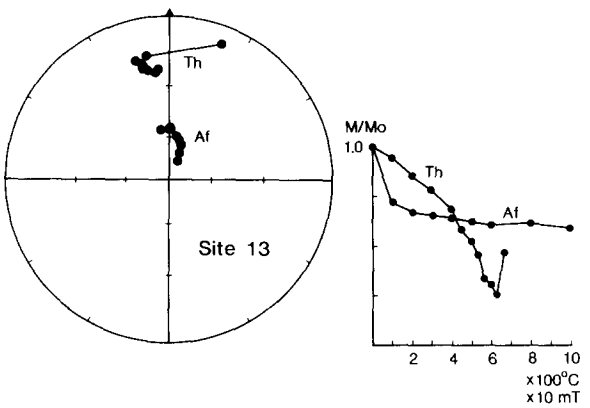


Fig. 5. Examples of alternating field and thermal demagnetization of specimens from the Båsnæring Fm., site 13. Conventions as in Fig. 2.

sites, samples were collected in red sandstones of the Hestmannen Member, the uppermost unit of the formation. A single characteristic component carried by haematite was isolated in the Båsnæring rocks; it has a shallow northerly direction (Fig. 8).

The improvements in α_{95} and k after tilt correction for the seven Båsnæring sites indicate that the characteristic magnetization in these rocks is pretilt in age. Before rotation to the palaeohorizontal, the α_{95} for the average of the Båsnæring sites was 27.4°, after rotation it was 10.6°. The corresponding data for k were 3.7 and 24.9.

Red mudstones from sites 19 and 20 of the Båtsfjord Formation yielded easterly directions with steep to intermediate positive inclinations. Site 21 gave a northerly direction with a shallow inclination. All sites are situated in the lower part of the formation, close to the contact with the Båsnæring Formation.

6. Discussion

From the results obtained it is possible to discern two groups of directions: one with steep positive inclinations (A; sites 5, 7, 9–11, 19 and 20) and another with a shallow northern direction (B; sites 2–4, 12–19, 21). Site 8 has a position intermediate between these groups and sites 1 and 6 are well outside the groups. The scatter in directions both between and within some of the sites is large and the lack of stratigraphical

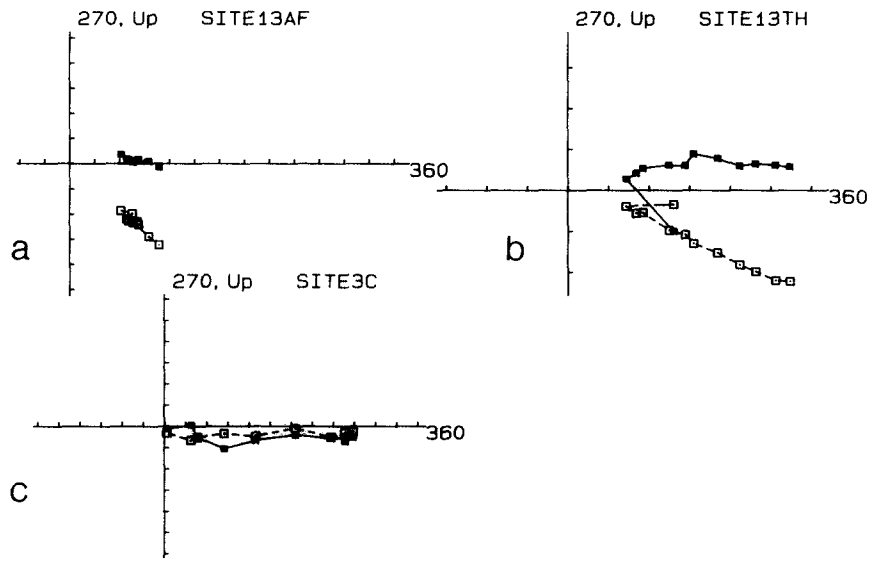


Fig. 6. Orthogonal vector projections of specimens from the Båsnæring Formation. Conventions as in Fig. 4. Units on axis (a), 0.5 mA/m, on (b) and (c), 1 mA/m.

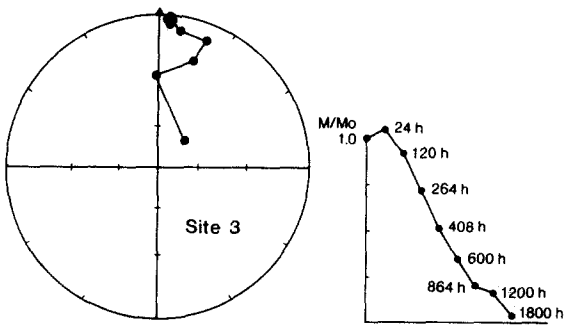


Fig. 7. Example of chemical demagnetization of a specimen from the Vadsø Group using 8 *N* hydrochloric acid. Conventions as in Fig. 2.

correlation between directions implies that some of the characteristic magnetization obtained are secondary. The good preservation of the microfossils (Vidal, 1981; Vidal and Siedlecka, 1983) and the low degree of metamorphism together with the lack of tectonic disturbances within the Vadsø Group indicate that the magnetization is of diagenetic origin.

6.1. Palaeopoles

Palaeopoles for the sites and formations studied are given in Table 3 and Fig. 9. In addition,

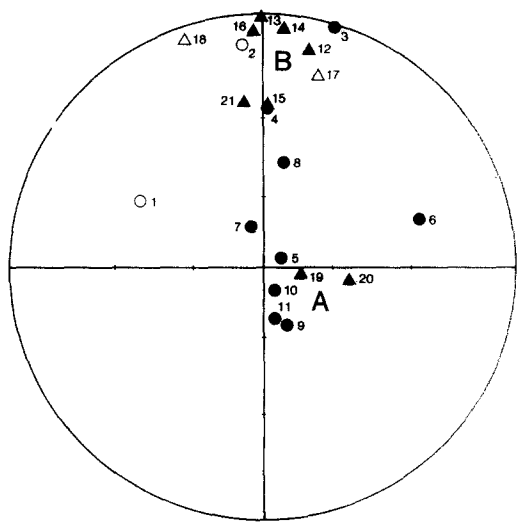


Fig. 8. Site mean directions from Table 2. Conventions as in Fig. 2. Circles/dots denote the Vadsø Group, triangles the Barents Sea Group.

poles from elsewhere in Fennoscandia are listed for comparison. The latter comprise poles from southern Scandinavia that form the younger part of the Sveconorwegian Loop of the Fennoscandian apparent polar wander path (APWP; Pesonen et al., 1989; Elming et al., 1993) and poles

Table 3
Late Riphean–Early Vendian palaeopoles from Fennoscandia

Rock unit	N	$P_{lat.}$ (°N)	$P_{long.}$ (°E)	d_p (°)	d_m (°)	Age (Ma)	Reference
Vadsø Group							
1. Veidnesbotn Fm.	6	−8	266	8	14		This paper
2. Klubbnes Fm.	5	16	214	8	16	809	This paper
3. Fugleberg Fm.	8	20	192	4	8		This paper
4. Fugleberg Fm.	7	34	209	8	14		This paper
5. Fugleberg Fm.	8	70	77	17	18		This paper
6. Fugleberg Fm.	8	18	131	9	17		This paper
7. Andersby Fm.	12	74	235	22	25		This paper
8. Andersby Fm.	10	47	193	12	19		This paper
9. Golneselv Fm.	4	26	49	26	33?		This paper
10. Golneselv Fm.	5	52	42	19	20		This paper
11. Ekkerøy Fm.	21	30	40	9	10		This paper
Barents Sea Group							
12. Båsnæring Fm.	20	24	204	3	7		This paper
13. Båsnæring Fm.	20	20	209	3	6		This paper
14. Båsnæring Fm.	1	22	205		–		This paper
15. Båsnæring Fm.	5	33	217				This paper
16. Båsnæring Fm.	1	22	212	–	–		This paper
17. Båsnæring Fm.	4	11	194	13	26		This paper
18. Båsnæring Fm.	5	22	230	7	13		This paper
19. Båtsfjord Fm.	11	52	88	10	16		This paper
20. Båtsfjord Fm.	1	29	100		–		This paper
21. Båtsfjord Fm.	7	32	219	11	20		This paper
Kola Peninsula							
31. Sandstones and carbonates Sredniy Peninsula	186	53	82	5	6	~620	Khramov, 1989
32. Dolerites, Sredniy Peninsula	12	52	52	17	17?	~600	Khramov, 1986
33. Grey beds, Kildanskaja Series	99	50	84	6	7	~600	Khramov, 1986
34. Kildanskaja Series	178	50	86	5	5	~600	Khramov, 1986
35. Grey beds, Kildin Island	67	49	94	8	9	~600	Khramov, 1986
36. Grey carbonates and terrig, rocks, Kildin Peninsula	61	51	98	7	9	~600 ~600	Khramov, 1989
37. Red beds, south Kildinskaya	15	38	191	7	11	~750	Khramov, 1986
38. Sandstones, marls, Kola Penins.	9	18	209	5	10	~750	Khramov, 1984
39. Red beds, Poropeonskaya Series	10	25	199	6	11	~750	Khramov, 1986
40. Poropelonkaya Series	8	39	199	7	13	~750	Khramov, 1986
41. Red beds, Kildinskaya Series	5	30	195	6	12	~750	Khramov, 1986
42. Red beds, Karujarvi Series	20	20	197	3	6	~750	Khramov, 1986
43. Red beds, Chernorechenskaya Series	40	29	189	2	4	~750	Khramov, 1986
South Sweden and Båtsfjord dykes							
51. Tärnö dolerite	5	0?	237	4	6	871	Bylund, 1992
52. Bräkne–Hoby dolerite	7	22	252	3	6	975–844	Bylund, 1992
53. Fäjö dolerite	6	23	249	3	5	935–800	Bylund, 1992
54. Visingsö Beds	11	0	258	9	16	>727	Bylund. prel. data
55. Båtsfjord dykes, Varanger	6	41	108	5	7	640	Kjode et al., 1978
Mean group B	11	23	209		9		This paper

N=number of samples used for calculation, $P_{lat.}$=latitude of the calculated palaeopole, $P_{long.}$=the calculated longitude of the palaeopole. d_p and d_m=the semi-axes of the error about the pole at the 95% probability level.

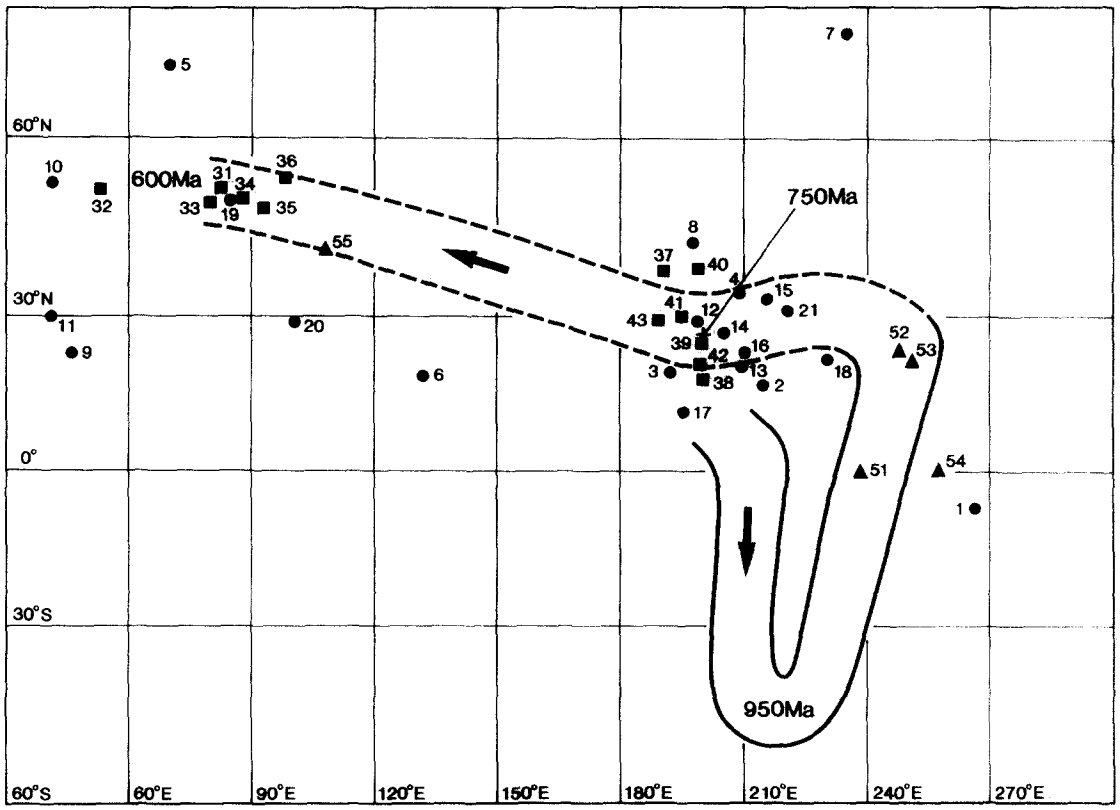


Fig. 9. The Fennoscandian apparent polar wander path sketched after Pesonen et al. (1989) and Elming et al. (1993). The path covers the approximate time interval 1050–600 Ma and starts with the Sveconorwegian Loop. Palaeopoles refer to Table 3. Dots denote palaeopoles from this paper, squares denote palaeopoles from the Kola Peninsula, and the triangles denote palaeopoles from south Sweden and the Båtsfjord dykes. The cross is the mean Group B pole.

from the northern part of the Kola Peninsula listed in Khramov 1984, 1986, 1989. The Kola palaeopoles are from sedimentary rocks and dolerites considered to be of Late Riphean and Vendian age (Siedlecka, 1975; Khramov, 1984, 1986, 1989) and situated between 69° and 70°N and between 32° and 34°E, ~100 km east of the Varanger Peninsula (K in inset Fig. 1).

The magnetic direction of the Veidnesbotn Formation of the Vadsø Group (entry 1 in Table 3) does not fit with Groups A or B. Its palaeopole is in the vicinity of the 850 Ma part of the APWP and the Visingsö Group palaeopole (entry 54, Table 2; Bylund, preliminary data). The Visingsö Group pole is based on data from the lower formation of the Visingsö Group of south Sweden (cross in inset, Fig. 1). The uppermost formation of the Visingsö Group has been dated at 727±37 Ma by Rb–Sr whole-rock analysis (Bonhomme and Welin, 1983). The Veidnesbotn Formation is older than 810±90 Ma (Rb–Sr whole rock), the age given for the Klubbnes Formation (Pringle, 1973). The pole positions suggest an age around 850–800 Ma for the Veidnesbotn Formation and the Middle Visingsö Group palaeopoles.

Group A palaeopoles are scattered but situated in the vicinity of the 600 Ma part of the Fennoscandian APWP. This implies a Vendian or post-Vendian remagnetization of these rocks.

Vidal (1981) and Vidal and Siedlecka (1983) considered the transition between the Upper Riphean and the Vendian to be situated in the Ekkerøy and Båtsfjord Formations (based on bios-

tratigraphic correlations) and implied that the Båtsfjord Formation palaeopoles 19 and 20 may represent a primary magnetization. However, this interpretation is contradicted by the Group B direction found in the Båtsfjord Formation, entry 21. Shipunov (1988) studied some of the rock units from the Kola Peninsula that are listed in Table 3 and plotted in the 600 Ma area in Fig. 9. He showed that their magnetization is post-folding and probably due to remagnetization caused by a 600 Ma old dolerite dyke. The dyke direction is decl. 88° and incl. 76°, which is comparable with the decl. 84° and incl. 71° for the 640 Ma old Båtsfjord dykes (Kjøde et al., 1978). Torsvik et al. (1990b), on the other hand, suggested that the Båtsfjord dykes became remagnetized during the Early Ordovician Finnmarkian event of the Caledonian Orogeny.

The B-group palaeopoles is situated in the same area as the 750 Ma palaeopoles from the Kola Peninsula (Table 3; Fig. 9) and it can be assumed that their magnetization is of Late Riphean age and thus in agreement with ages given by the microfossil determinations (Vidal, 1981; Vidal and Siedlecka, 1983). Rb–Sr whole-rock dating on cleaved mudstones from the Kongsfjord Formation suggests that the folding of the Barents Sea Group rocks occurred at 520 ± 47 Ma (Taylor and Pickering, 1981); an age that presents an upper limit for the magnetization of the Båsnæring Formation acquired before folding. The lower limit is given by the well dated (850 Ma) late Sveconorwegian palaeopoles from southern Sweden, 52 and 53 (Fig. 9).

The magnetization of the Varanger successions studied is post-depositional and diagenetic. Timing of the magnetization process is thus uncertain, as sediments could be magnetized just after deposition or later during uplift and change in environment up to recent time. Magnetization also depends on the original composition of the rocks. A complicating factor can be reversals of the ambient geomagnetic field during diagenesis, causing scatter in directions. These reasons can explain the scatter in data from the Varanger sequences and the presence of both Group A and B directions in the same formations. Examples of this are found in the Fugleberget and Båtsfjord Formations.

The shallow inclination of the B-group directions indicates that the Fennoscandian Shield was situated close to the equator during Late Riphean–Early Vendian times. The position of the Shield at ~850 and 750 Ma, based on data from the Varanger Peninsula sedimentary successions is shown in Fig. 10.

6.2. *The Trollfjord–Komagelv Fault*

The palaeomagnetic results available from the Vadsø and Barents Sea Groups do not confirm

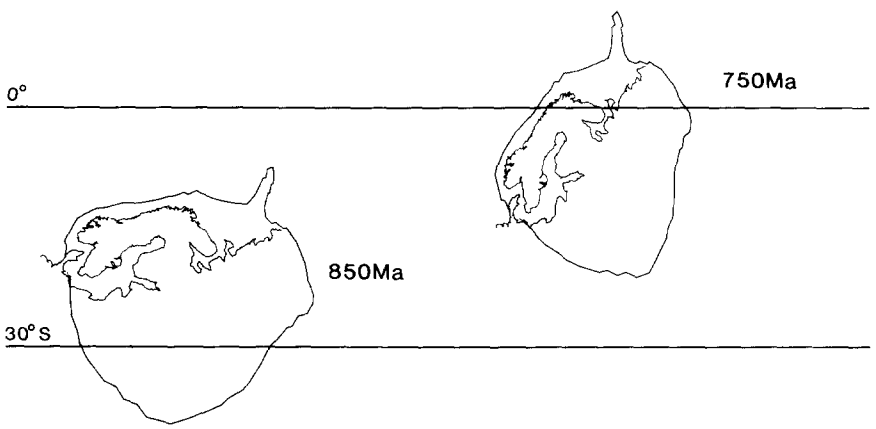


Fig. 10. The latitudinal position of Baltica according to the B-group data (~750 Ma) and at the end of the Sveconorwegian Orogeny at ~850 Ma. The longitudinal position is arbitrary. The reconstruction is made with the program GMap 4 (Torsvik et al., 1990a).

movements along the Trollfjord–Komagelv Fault of the magnitude suggested by Kjøde et al. (1978) and Pesonen et al. (1989). Group B palaeopoles of Late Riphean age from both sides of the fault occur in the same cluster with no significant difference in position (Fig. 9). It is evident that previously suggested movements of 1000 to 500 km are not confirmed by present palaeomagnetic data. However, the existence of the fault and the major lithologic contrasts across this major tectonic feature suggest substantial tectonic displacements. Nevertheless, these displacements should be within the confidence errors of the palaeomagnetic measurements, ~ 10–15°, which allow movements of up to 250 km along the fault using the rotation pole and data given by Kjøde et al. (1978). This movement is in somewhat better agreement with the dislocation of 50 ± 50 km suggested by Townsend et al. (1990) and Rice and Townsend (1991).

7. Conclusions

The palaeomagnetic study of sedimentary rocks of the Vadsø and Barents Sea Groups show that several sequences contain characteristic magnetization of diagenetic origin of Late Riphean age. In addition, there are sequences that yield palaeopoles in disaccordance with their inferred ages due to partial or complete remagnetization. The age of this Group A remagnetization is uncertain but is in the range Late Vendian to Early Ordovician.

The palaeomagnetic data obtained from sequences on both sides of the Trollfjord–Komagelv Fault indicate that the movement of the Barents Sea Region relative to the South Varanger Region is less than 250 km, substantially less than suggested earlier.

Acknowledgements

Constructive comments on the first draft by Gonzalo Vidal and Ian Snowball are appreciated. Gonzalo Vidal initiated the Varanger studies and has been an excellent companion in the field. Many thanks are due to Anna Siedlecka and Stan Siedlecki who introduced us to the geology of the Varanger Peninsula and were superb guides. Britt Nyberg drafted the figures. The Swedish Natural Science Research Council financed the work.

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